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A validation of the MACS toolbox for EEG data analysis and second-level fMRI

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Introduction

The MACS toolbox [1,2,3] is an SPM extension that allows for model assessment, comparison and selection of first-level general linear models (GLMs) for functional magnetic resonance imaging (fMRI) data. Since its publication, questions have been raised whether MACS also enables comparison between GLMs for EEG – using e.g. different trial categorizations or different parametric regressors – or second-level GLMs for fMRI – using e.g. different covariates such as sex, age or handedness [4]. Here, we introduce a simple extension of the toolbox code that allows to perform such non-standard analyses [8].

Methods

The default model quality criterion in the MACS toolbox, the so-called cross-validated log model evidence (cvLME), is based on a partition of the scans described by each GLM

$$\mathrm{cvLME}(m) = \sum_{i=1}^{S} \log \int p(y_i|\theta,m)\, p\left(\theta \middle| \bigcup_{j\neq i} y_j\right) \mathrm{d}\theta$$

where y_i is the i-th part of the data (e.g. 1st recording session or 1st half of scans) and S is the number of data subsets.

For second-level GLMs, the design matrix is split in two parts, but it must be ensured that observations from each cell – e.g. in an ANOVA – are included in each of the two parts. For this reason, the data y are randomly split into mutually exclusive and collectively exhaustive subsets $\tilde{y}_1$ and $\tilde{y}_2$ that have the same size:

$$\tilde{y}_1, \tilde{y}_2 \subset y, \;\; \tilde{y}_1 \cap \tilde{y}_2 = \emptyset, \;\; \tilde{y}_1 \cup \tilde{y}_2 = y$$
$$p(y_i \in \tilde{y}_1) = p(y_i \in \tilde{y}_2) = 1/2\,, \;\; i = 1, \ldots, n$$
$$n_1 = n_2 = n/2\,, \;\; n_1 + n_2 = n$$

In order to ensure the same splitting for different models describing the same data, MATLAB's random number generator (`rng`) is seeded with the number of scans. This procedure is available as the new routine »`MA_cvLME_other`« in the MACS toolbox [8].

After cvLME maps have been generated, models are compared using log Bayes factor (LBF) maps by simply subtracting two cvLME maps from each other [3]:

$$\mathrm{LBF}(m_1, m_2) = \mathrm{cvLME}(m_1) - \mathrm{cvLME}(m_2)$$

An LBF larger than three is considered "strong evidence" in favor of a model in Bayesian statistics [9]:

$$\mathrm{LBF}(m_1, m_2) > 3 \Rightarrow m_1 \text{ better than } m_2$$

Free Space / Art Corner

Results: first-level EEG

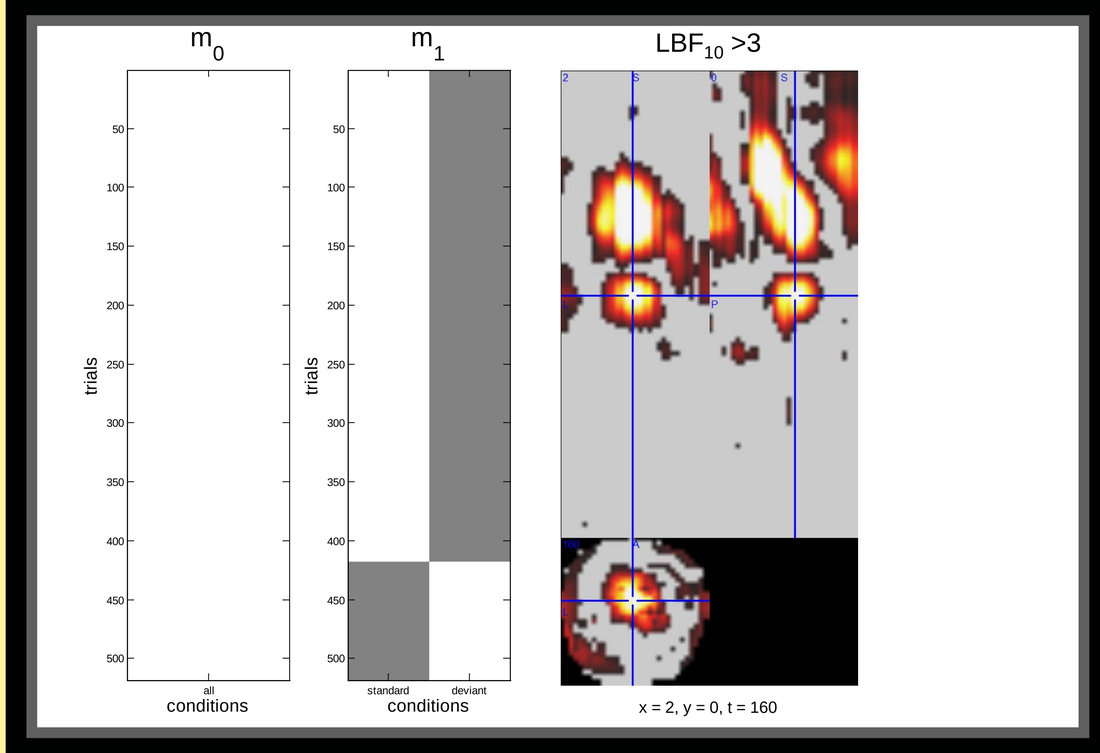

Figure 1. *EEG data analysis.* The model distinguishing standard and deviant tones outperforms the model not doing so in a fronto-polar region close to 160 ms, demonstrating mismatch negativity.

For EEG validation, we use the single-subject auditory mismatch negativity data set available from the SPM website, acquired by [5] and preprocessed according to [7]. In this experiment, 480 standard and 120 deviant tones were presented from which some were eliminated during artifact detection. For sensor space analysis, two models were specified, one assuming a difference between the two conditions (m_1) and the other assuming no such difference (m_0). Consequently, when there is a difference between standard and deviant tones, m_1 should outperform m_0. When thresholding at $\mathrm{LBF} > 3$, we indeed observe such a model preference in a fronto-central region close to 160 ms after the stimulus (see Figure 1). This replicates the famous mismatch negativity in EEG and is consistent with classical analysis of the data [7].

Results: second-level fMRI

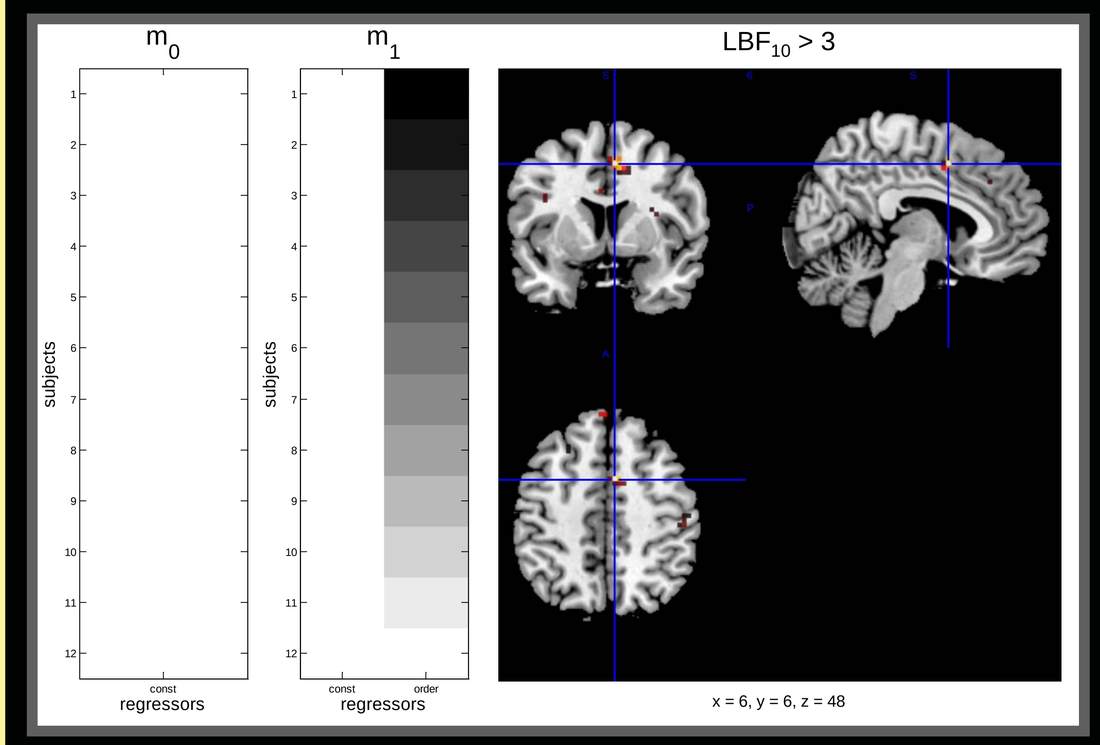

Figure 2. *fMRI data analysis.* The model including subject order as a covariate in a one-sample t-test outperforms the model not doing so in a cingular region, indicating that this covariate should be included into the model when testing for an overall effect of faces in this region.

For fMRI validation, we use the multi-subject face repetition priming data set available from the SPM website, acquired by [6] and preprocessed according to [7]. In this experiment, faces belonging to four different categories were presented and the contrast "faces versus baseline" was prepared for group statistics. For second-level analysis, two models were specified, one using the mean over subjects and subject order as a covariate (m_1) and the other having no covariate (m_0). Since we do not expect subject order to have an effect on the data, there should be no or only few voxels preferring m_1 over m_0. When thresholding at $\mathrm{LBF} > 3$, we indeed only observe a small cingular region in which order should be included into the model (see Figure 2). This is consistent with an uncorrected classical analysis of the data.

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Discussion

Using a first-level EEG analysis giving a positive result and a second-level fMRI analysis yielding a largely negative result, we have validated the MACS toolbox for model comparison of non-first-level GLMs. In SPM's factorial design specification, there are two possible violations of independent and identically distributed (i.i.d.) errors, namely (i) dependences between different levels of a factor (non-independence) and (ii) unequal variances between those levels (heteroscedasticity). While heteroscedasticity is unproblematic for the random split-half procedure described above, the effect of non-independence on the cvLME for second-level GLMs needs to be further investigated. In cases of uncertainty, the MACS toolbox also allows for classical model selection using AIC or BIC [3].

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